Supplemental Material for
“Normal-Form Resolution of a Free-Boundary MOTS Saddle-Node in Dynamical Five-Dimensional Gravity”

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1 Purpose and evidentiary scope

This supplement gives the definitions, acceptance criteria, extended numerical results, negative studies, and reproducibility structure underlying the main article. The central inference is deliberately bounded: an independently constructed G9/G10/G11 parent sequence evolves from no accepted upper-brane-capped marginally outer trapped surface (MOTS) to two opposite-stability branches, and pseudo-arclength continuation resolves their common critical surface. An isolated near-critical principal mode accompanies the bracketed zero crossing, the endpoint-eliminated discrete adjoint transversality and quadratic coefficients are nonzero, and the invariant area separation follows a one-half power. This establishes a finite-resolution numerical free-boundary MOTS saddle-node in the $SO(3)$-symmetric sector of the specified stabilized compact two-brane model. An independent G10 half-timestep evolution and continuation strongly constrains the shared temporal-bias concern under prospective comparison criteria: the critical-time brackets overlap, while the near-critical geometry and normal-form diagnostics agree. The saddle-node mechanism itself is established theory [1]; it is not claimed as a new mechanism.

The result is numerical and quasi-local. It is not an event-horizon reconstruction, a proof excluding every possible initial marginal surface, a continuum theorem, or a nonlinear-stability theorem. It also does not determine the nonsymmetric nonprincipal spectrum. Those exclusions are part of the result's scope, not caveats to be silently relaxed.

The validation record is organized by scientific risk. Each principal test has four elements: the alternative explanation being tested, a criterion fixed before interpretation, the measured outcome, and the consequence for the claim. Operational details—hashes, complete runner histories, environment receipts, and restart records—belong to the reproducibility archive rather than to the scientific argument.

The novelty claim is correspondingly narrow. Cigar-shaped apparent horizons in braneworld initial data and dynamically formed brane-intersecting horizons in RSII precede this work [2, 3], and MOTS pair creation is already classified as a saddle-node bifurcation [1]. Free-boundary MOTS stability, rigidity, and higher-dimensional structure likewise have an established mathematical literature [4, 5, 6, 7, 8]. Free-boundary minimal annuli also provide a neighboring static fold and Jacobi–Robin degeneration result [9]; that setting does not evaluate the dynamical MOTS adjoint transversality and quadratic coefficients tested here. The unresolved issue addressed here is whether the complete local fold hypotheses are realized when a physical contact condition enters the MOTS boundary-value map and the ambient geometry is supplied by a nonlinear evolution.

The contribution is therefore the closed numerical chain: the physical Robin domain, branch connection, simple principal zero mode, nonzero endpoint-eliminated discrete adjoint projections, invariant square-root separation, three-grid spatial closure, and an independent half-timestep comparison. The stabilized compact braneworld supplies the concrete five-dimensional realization. One such realization establishes that the physical boundary does not obstruct the local fold structure in this problem; it does not establish genericity, a universal theorem, or a new bifurcation mechanism.

2 Model, fields, and conventions

The bulk model is five-dimensional Einstein gravity with $\Lambda_5 = -6/\ell^2$ on the doubled $S^1/\mathbb{Z}_2$ cover. The numerical domain is the fundamental compact interval with fixed branes at its endpoints. The matter sector contains a Goldberger–Wise stabilizer Φ and an independent collapse scalar χ [10, 11, 12, 13]:

$$
\begin{aligned}
S = \frac{1}{2\kappa_5^2}\int \mathrm{d}^5x\sqrt{-g}\,(R-2\Lambda_5) - \sum_i \int_{\Sigma_i} \mathrm{d}^4x\sqrt{-h_i}\,[\lambda_i + U_i(\Phi)] \\
-\frac{1}{2}\int \mathrm{d}^5x\sqrt{-g}\left[(\nabla\Phi)^2 + m_\Phi^2\Phi^2 + (\nabla\chi)^2 + m_\chi^2\chi^2\right], \qquad (1)
\end{aligned}
$$

with $U_i(\Phi) = \gamma_i(\Phi - v_i)^2/2$. The induced metric obeys the oriented Israel conditions, the stabilizer obeys the action-derived scalar wall condition, and χ obeys reflecting Neumann data. The computation retains the stabilizer response; it does not integrate out the radion. The displayed action is the doubled-cover thin-shell action. An equivalent one-sided interval variational principle includes the appropriate Gibbons–Hawking–York terms and doubles the one-sided bulk contribution relative to each brane action [14, 15, 16]. With outward normals it yields

$$
K^{(B)}_{\mu\nu} = \frac{\kappa_5^2}{2}\left(S_{\mu\nu} - \frac{1}{3}S h_{\mu\nu}\right), \qquad n^a_{\mathrm{out}}\nabla_a\Phi = -\frac{1}{2}U_i'(\Phi), \qquad (2)
$$

which is the sign and factor convention implemented at both interval walls.

We impose $SO(3)$ symmetry in the three noncompact spatial directions while retaining dependence on areal radius r and compact coordinate z. Initial data are momentarily time symmetric,

$$
K_{ij} = 0, \qquad \Pi_\Phi = 0, \qquad \Pi_\chi = 0, \qquad (3)
$$

and contain a smooth reflection-adjusted scalar pulse localized near the upper brane. The primary datum is the family member labelled $A = 7.90$. The label is a numerical family coordinate, not an invariant critical parameter.

Table 1 gives the dimensionless parameters needed to identify the reported family. We use $\ell = \kappa_5 = 1$, so the compact coordinate spans $z \in [1, e]$. The load-bearing repaired G9/G10/G11 line uses $r/\ell \in [0, 10]$. The superseded pre-repair G7/G8 trajectory, retained only as a historical record below, used $r/\ell \in [0, 8]$. The stabilizer targets are $v_0 = \sqrt{0.01}$ and $v_1 = v_0 e^{-0.1}$, and $m_\Phi^2 = 0.1(4.1) = 0.41$. The wall tensions are retuned consistently with the finite-stiffness background solve rather than held at their unstabilized values.

The background values in Table 1 are the sealed binary64 parent values; the last quoted digits of β_1, the upper-wall tension, and the proper separation vary below 7×10^{-13}, 5×10^{-13}, and 3×10^{-10},

Table 1: Primary dimensionless model and discretization parameters.

Quantity	Value
ℓ, κ_5, compact separation d/ℓ	1, 1, 1
ϵ, Goldberger–Wise backreaction, m_Φ^2	0.1, 0.01, 0.41
wall stiffness $\gamma\ell$	20
wall targets v_0, v_1	0.1, 0.0904837418036
retuned bare tensions λ_0, λ_1	5.9999903768, −6.0528442777
background warp slopes β_0, β_1	1, 1.00880714399
background proper separation$/\ell$	0.9996389584
collapse-scalar mass m_χ^2	0
pulse amplitude A, radial width σ_r	7.90, 1
compact-log width σ_y, center y_c/d, images	0.2, 0.9, 5
historical pre-repair domain and grids	$R_{\max}/\ell = 8$; G7 81×121, G8 97×145
repaired critical-sequence domain	$R_{\max}/\ell = 10$
repaired critical grids	G9 113×211, G10 129×241, G11 145×271
critical and half-step $\Delta t/\ell$	3.125×10^{-5}, 1.5625×10^{-5}
background solve tolerance	10^{-10}

respectively, between the two archived parent resolutions. These numerical values, rather than only their analytic target formulas, are included in the replay package.

For a spatial-slice unit normal u^a and outward surface normal s^a, the outgoing null normal is $\ell_+^a = u^a + s^a$ and

$$\theta_+ = D_i s^i + K_{ij} s^i s^j - K. \tag{4}$$

A MOTS satisfies $\theta_+ = 0$. An upper-brane-capped surface closes regularly on the axis and meets the upper brane orthogonally. Orthogonal contact with the support hypersurface is the free-boundary condition used in the cited literature, and it is precisely the brane endpoint condition imposed here. The cap is therefore a brane-terminating free-boundary surface; free-boundary and capillary MOTS stability have an independent mathematical literature [4, 5, 6, 7, 8], while higher-dimensional MOTSs have also been treated directly [17]. Linear stability is defined by

$$Lf = \delta_{fs}\theta_+. \tag{5}$$

In the convention used here, a nonnegative principal eigenvalue is outward-stable. The evolved-slice operator is treated as non-self-adjoint. The interpretation follows the local MOTS-stability framework of Refs. [18, 19, 20, 21]. For the connected cap, the scalar stability operator is strongly elliptic, and its Robin domain commutes with $SO(3)$. Free-boundary principal-eigenvalue theory supplies a positive eigenfunction that is unique up to scale [4, 8]. A group transform is therefore the same normalized principal eigenfunction. The symmetric computation captures the full principal eigenvalue and its simple kernel, although it does not determine the nonsymmetric nonprincipal spectrum.

The perturbation class is not left implicit. Writing a nearby cap as $\rho(\vartheta) + \delta\rho(\vartheta)$, smooth-axis closure and preservation of orthogonal wall contact impose

$$\partial_\vartheta \delta\rho = 0 \quad \text{at} \quad \vartheta = 0, \pi/2. \tag{6}$$

The operator acts on physical normal displacement $f = w\delta\rho$, where $w = s_A e_\rho^A > 0$. Hence the implemented condition is $\partial_\vartheta(f/w) = 0$, a homogeneous Robin condition for f when w is nonconstant.

The endpoints are eliminated with seven-point local finite differences before the unsymmetrized right-eigenvalue problem is solved. This boundary treatment is checked by the manufactured Robin spectrum, endpoint defects below 10^{-10}, and the 49/65/81-node sequences. No alternative stencil-width study is claimed.

3 Initial-data and search protocol

3.1 Bounded initial search

The load-bearing absence datum is the repaired G9/G10/G11 search at $t/\ell = 0.001$ described in Sec. 8.1; it is followed by exactly two admitted surfaces at $t/\ell = 0.0015$ on those same parent lines. The earlier searches summarized in this subsection were performed during the superseded pre-repair study and are preserved only to document how the repaired calculation was motivated. They are not independent confirmation of the repaired result. Within that historical record, "no initial MOTS" means "no accepted surface in the prespecified bounded search," not an analytic nonexistence theorem. The principal initial class consists of C^2, axisymmetric, star-shaped upper-brane-capped polar graphs

$$z = z_B - \rho(\vartheta)\cos\vartheta, \qquad r = \rho(\vartheta)\sin\vartheta, \tag{7}$$

with $0 \le \vartheta \le \pi/2$, $\rho'(0) = \rho'(\pi/2) = 0$, and $0.10 \le \rho(\vartheta) \le 1.67$. Separate bounded searches cover opposite-side capped, closed-star-shaped, and interval-spanning candidates. Acceptance requires finite geometry, admissible domain and endpoint behavior, a residual below the fixed tolerance, and repeated recovery under the specified seed families.

For the historical $A = 7.90$ datum, the initial upper-brane-capped searches return no accepted surface. The same is true for the five-point R8 sample $A = 7.84, 7.86, 7.88, 7.90, 7.92$. An adverse $A = 7.94$ control recovers its known initial pair. On each separately constraint-solved G9 and G10 initial slice, a regular-axis shooting audit samples 2,049 uniformly spaced axis radii across $[0.10, 1.67]$. The independently minimized brane residual stays positive, 0.0280624085 on G9 and 0.0282193410 on G10, under twelve cutoff, tolerance, and step variations. The same implementation resolves both nearby $A = 7.94$ control roots to residuals of order 10^{-13}. The grade remains a strong floating-point no-root result rather than an interval or degree proof: an arbitrarily narrow unsampled root island is not mathematically excluded.

3.2 Surface representations

On the historical pre-repair trajectory, the primary evolved-surface representation uses the fixed twelve-seed bank $1.15, 1.20, \ldots, 1.70$. Every seed is solved at 40 and 48 cosine modes on 257 collocation points, with expansion target 5×10^{-4}. A candidate requires two-mode convergence in-domain below condition number 10^5, full profile and endpoint agreement within 0.2%, and an independent 513-point expansion below 0.002; signatures within 0.005 are deduplicated. The confirmation representation is an independently parameterized local boundary-value problem. Acceptance requires agreement in branch count, endpoint geometry, and proper cap area. Both representations operate on the same evolved spacetime, so their agreement constrains detector and parameterization error but is not an independent evolution.

The 10^5 condition-number ceiling belongs only to that historical spectral detector. The load-bearing repaired search and critical continuation below use the distinct local second-order BVP and does not call or inherit that ceiling; its admission uses the local residual, endpoint slope, physical domain, and independent expansion cross-check. Accordingly, proximity to the principal zero cannot remove a continuation point through the spectral-detector condition gate. No critical condition-number trace was archived, and none is used as evidence for the saddle-node classification.

The BVP detector solves the local second-order cap equation with $\rho'(0) = \rho'(\pi/2) = 0$ from the same twelve scalar seeds. Admission requires local interior expansion below 2×10^{-4}, endpoint-slope error below 2×10^{-4}, and a cross-check by the primary expansion evaluator below 0.002. Its flat-slice control reproduces the analytic marginal half-sphere radius 1.25 to 3.91×10^{-11} relatively and gives an independent expansion maximum 3.66×10^{-6}.

In the historical pre-repair record, at $t = 0.000625$, 0.001, and 0.004, each G7/G8 grid yields exactly two accepted branches, with six seed recoveries per branch. The largest shooting/BVP endpoint mismatch is 0.04764% and the largest cross-grid BVP endpoint transfer is 0.23205%.

4 Evolution and ordered acceptance gates

The Einstein–scalar system is evolved in a regular $SO(3)$-reduced generalized-harmonic formulation [22]. Compact-wall rows own the wall conditions; axis regularity owns interior-axis rows; wall–axis and wall–outer corners use an explicitly staged ownership policy. The canonical policy is the scientific path, while an independent wall-owner policy is retained as a control.

The ordered gate structure prevents a later geometric observation from overriding an earlier invalid computation:

1. runtime, source, and input identity;
2. finite values, signature, and physical admissibility;
3. compact-wall, axis, corner, and ownership identities;
4. repeated deterministic replay;
5. temporal refinement and endpoint consistency;
6. spatial field and physical-tensor closure;
7. horizon count, geometry, and stability interpretation.

An audit or determinism failure is not relabelled as a scientific negative result. A resource stop carries no scientific inference.

5 Formation, geometry, and stability

5.1 Free-boundary stability condition

Let ϱ be the outward unit normal to the brane within the spatial slice, N the unit normal to the MOTS, and ν the outward conormal to its boundary. Orthogonal contact gives $\nu = \varrho$ and $g(N, \varrho) = 0$. For a physical normal variation fN, varying this orthogonality relation gives the free-boundary operator

$$Bf = \nabla_\nu f - \mathrm{II}^{\partial M}(N, N) f = 0. \tag{8}$$

Thus the brane-extrinsic-curvature term is part of the physical problem.

For the implemented upper-brane cap, the compact endpoint lies on the upper wall $z = e$. The retained interval lies toward decreasing z, so its outward wall normal is $\varrho = h_{zz}^{-1/2} \partial_z$. We choose the outward cap conormal $\nu = \varrho$, the MOTS normal toward increasing areal radius, $N = h_{rr}^{-1/2} \partial_r$, and $\mathrm{II}^{\partial M}(X, Y) = g(\nabla_X \varrho, Y)$. Wall parity gives $h_{zr} = 0$. With these fixed orientations, the physical normal amplitude associated with a coordinate graph displacement $\delta\rho$ is $f = w\delta\rho$ with $w = \sqrt{h_{rr}}$. The same geometry gives

$$\mathrm{II}^{\partial M}(N, N) = \frac{1}{2} \nu(\log h_{rr}) = \nu(\log w), \qquad Bf = w\nu(\delta\rho). \tag{9}$$

Consequently the implemented endpoint condition $\partial_\vartheta \delta\rho = 0$, equivalently $\partial_\vartheta(f/w) = 0$, is the coordinate reduction of the physical Robin condition; it does not omit the brane-curvature contribution. The zero derivative at the axis is separately the smooth-axis regularity condition.

5.2 Three-grid formation comparison

The primary repaired critical line is independently constructed on G9 113×211, G10 129×241, and G11 145×271. The prescribed surface search accepts no branch at $t/\ell = 0.001$ and exactly two at $t/\ell = 0.0015$ on every grid. All endpoint, proper-geometry, principal-sign, operator-resolution, and Robin-boundary gates pass. The direct observables are listed in Table 2.

Table 2: Three-grid repaired-parent surface observables at $t/\ell = 0.0015$.

grid	$\mathcal{A}_{\rm in}$	$\lambda_{\rm in}$	$\mathcal{A}_{\rm out}$	$\lambda_{\rm out}$
G9	40.67520320	−0.25894888	40.72845794	+0.29347891
G10	40.67520099	−0.25895908	40.72845823	+0.29348687
G11	40.67519977	−0.25896247	40.72845818	+0.29348984

For G9–G10, the maximum endpoint, geometry, and relative-eigenvalue differences are 1.022×10^{-6}, 1.764×10^{-7}, and 3.938×10^{-5}, respectively. For G10–G11 they are 6.529×10^{-7}, 1.177×10^{-7}, and 1.311×10^{-5}. Thus the repaired line's earlier absence at $t/\ell = 0.001$ is a pre-formation datum, not a permanent failure to form the pair.

5.3 Pseudo-arclength continuation and zero bracket

Starting from the outer branch at $t/\ell = 0.0015$, time is promoted to an unknown of the local cap BVP and an integral pseudo-arclength condition is added [23]. The spacetime state between exact RK2 endpoints uses the fixed dense extension

$$y(\theta) = y_n + \Delta t[(\theta - \theta^2)k_1 + \theta^2 k_2], \qquad \theta = (t - t_n)/\Delta t. \tag{10}$$

Every replayed endpoint is bitwise identical to the stored evolution. Each accepted continuation point starts from arclength step 1/64 and permits only the prespecified dyadic backoff through 1/4096; zero-bracket refinement has the technical floor 1/65536. The augmented BVP uses 121 initial angular nodes, relative tolerance 2×10^{-6}, at most 6,000 adaptive nodes, and a 501-node dense output. An accepted corrector must also have absolute arclength residual at most 2×10^{-5} before the fixed-time surface is re-solved and passed through the inherited local and cross-evaluator gates.

The principal eigenvalue first changes sign and remains resolved under bracket refinement. Table 3 reports the final bracket and near-critical discrete data. The displayed λ_* and coefficients are evaluated at the smaller-magnitude endpoint of the final bracket rather than at an exact floating-point kernel.

The mode convention is exact. The Robin endpoint rows are first eliminated, so the stability matrix acts on physical normal displacement f at the interior uniform ϑ nodes. Its right eigenvector is phased so that the component of largest modulus is positive real and then scaled to $\|f_R\|_\infty = 1$. The left vector is the Euclidean conjugate-transpose eigenvector of that reduced matrix, not a mass-matrix adjoint or a separately discretized formal continuum adjoint. It is scaled afterward to $f_L^* f_R = 1$. Thus the reported unit overlap is a construction identity, not an additional nondegeneracy observation. Before rescaling, the code requires only that the overlap magnitude exceed 10^{-14};

Table 3: Three-grid critical-time estimate and near-critical discrete operator and normal-form data. $\mathcal{A}_{\rm nc}$ is evaluated at the selected bracket endpoint rather than an exact critical surface.

quantity	G9	G10	G11
t_*/ℓ	0.0011725421	0.0011725452	0.0011725359
time-bracket width	1.8426×10^{-7}	1.7177×10^{-7}	1.8667×10^{-7}
$\mathcal{A}_{\rm nc}$, one-sided	40.70345140	40.70336590	40.70344580
λ_*	-7.3854×10^{-4}	-1.6944×10^{-3}	-7.8666×10^{-4}
next reduced-sector eigenvalue	5.56598	5.56531	5.56595
imposed $f_L^* f_R$	1.0000000	1.0000000	1.0000000
near-critical $a = \langle f_L, \partial_t \theta_+ \rangle$	−20.76513	−20.76565	−20.76493
near-critical $b = \frac{1}{2}\langle f_L, \partial^2_{f_R} \theta_+ \rangle$	2.80717	2.80410	2.81060

that value was not archived. The endpoint behavior of f_R is restored through the same Neumann extension that implements the Robin graph condition. The near-critical coefficient magnitudes and individual signs are reproducible under this discrete orientation convention, while their nonvanishing is the normal-form conclusion. An admissible change of eigenvector normalization rescales a and b by finite nonzero factors; it changes their numerical magnitudes but cannot change whether either coefficient vanishes.

The near-critical principal mode is real and sign-definite on each grid. The 65–81-node principal differences lie between 3.59×10^{-5} and 3.78×10^{-5}; the doubled-step differences are approximately 4.78×10^{-4}. The next mode is more than five units away. Centered step halving changes a by at most 1.56×10^{-9} and b by at most 2.67×10^{-3} without changing either sign. Relative adjacent-grid transfers are at most 3.46×10^{-5} for a and 2.32×10^{-3} for b, far below the prospective 0.20 ceilings. The G9/G10/G11 sequences for t_*, $\mathcal{A}_{\rm nc}$, a, and b are all nonmonotone. Their full relative spreads are, respectively, 7.87×10^{-6}, 2.10×10^{-6}, 3.46×10^{-5}, and 2.31×10^{-3}. Consequently no real three-level Richardson order is defined and no continuum-extrapolated value or uncertainty is inferred. The evidence is small finite-resolution scatter plus the independent temporal comparison below. The isolated near-critical principal mode and bracketed zero crossing are consistent with a simple principal critical kernel. The roles of that kernel, the transversality coefficient, and the nonzero quadratic term are those of the standard fold normal form [24].

5.4 Invariant square-root test

Both branches are solved at the five prespecified offsets $t = t_* + (1, 2, 4, 8, 16)\Delta t/16$. A linear fit of $(\mathcal{A}_{\rm out} - \mathcal{A}_{\rm in})^2$ against time gives the results in Table 4. The fitted critical times agree with the continuation estimates within the frozen temporal criterion.

Table 4: Invariant proper-area scaling fits. The displayed R^2 is for the linear fit of $(\Delta\mathcal{A})^2$ against time; the exponent comes from a separate log–log fit using the critical time inferred by that linear fit.

grid	$t_{*,\rm fit}/\ell$	free exponent p	squared-fit R^2
G9	0.0011725485	0.4959125	0.9999801
G10	0.0011725343	0.4959672	0.9999804
G11	0.0011725265	0.4959590	0.9999803

All nine per-grid critical gates and both adjacent-grid aggregate gates pass.

The fit-sensitivity audit separates the two fits rather than treating their statistics as interchangeable. Holding t_* at the independently obtained continuation midpoint gives free exponents 0.49644, 0.49507, and 0.49518 on G9, G10, and G11. Omitting one offset at a time gives the ranges 0.49361–0.49956, 0.49187–0.49865, and 0.49201–0.49872. Fixing $p = 1/2$ and optimizing only the amplitude gives maximum relative residuals 0.00920, 0.01173, and 0.01153. These tests support the half-power classification while showing that the seventh decimal place of the original free exponent is not an uncertainty estimate.

5.5 Independent G10 half-timestep comparison

The three spatial grids use the same RK2 step and dense extension. A prospectively frozen G10 calculation therefore halved the step from 3.125×10^{-5} to 1.5625×10^{-5}, reconstructed the parent from $t/\ell = 0.001$ to 0.0015 in 32 steps, solved both endpoint anchors anew, replayed the complete half-step trajectory, and repeated every load-bearing continuation and critical calculation. Table 5 gives the result.

Table 5: G10 half-timestep comparison. The t_* entries are zero-bracket midpoints; their difference is not an uncertainty estimate. The $\mathcal{A}_{\rm nc}$, a, and b differences are relative, with $\mathcal{A}_{\rm nc}$ evaluated at the selected near-critical endpoint.

quantity	full step	half step	difference
t_*/ℓ	0.001172545169	0.001172544252	9.1732×10^{-10}
$\mathcal{A}_{\rm nc}$	40.70336590	40.70340030	8.4495×10^{-7}
a	−20.76564697	−20.76535233	1.4189×10^{-5}
b	2.80409588	2.80435486	9.2348×10^{-5}
free exponent p	0.495967	0.498057	—
squared-fit R^2	0.9999804	0.9999951	—
zero-bracket width	1.7177×10^{-7}	1.7918×10^{-7}	—

The full- and half-step zero brackets overlap. Their midpoint shift is 9.1732×10^{-10}, whereas their widths are 1.7177×10^{-7} and 1.7918×10^{-7}; the shift is not an uncertainty estimate for t_*. The half-step result retains the real sign-definite near-critical principal mode, reduced-sector spectral gap, nonzero coefficient signs, and area-scaling gate. Both independently evolved parents also pass the same finite-value, signature, constraint, wall, axis, and endpoint gates. This repeat is an indirect check against a systematic temporal-evolution error; it is not a standalone constraint-convergence study. The free exponent moves from 0.495967 to 0.498057, toward the expected one-half value. Together with the fit-sensitivity ranges above, that movement is consistent with the residual exponent offset being dominated by temporal-discretization bias rather than a fitting artifact, but two time steps do not define a temporal convergence order. The prospective limits were $\Delta t/64$ for the bracket-midpoint shift, 1% for the area, 20% for each coefficient, and $0.4 \le p \le 0.6$ in both runs. Every comparison passes, so the prescribed decision sequence does not call for a quarter-step run.

5.6 Historical pre-repair formation record

The following G7/G8 trajectory predates the canonical parity repair and is retained for provenance rather than counted as independent evidence for the repaired saddle-node. Both historical grids are empty in the bounded initial upper-brane-capped search and then contain two accepted branches in the common bracket

$$0.000500 < t_{\mathrm{form}} \leq 0.000625. \tag{11}$$

The pair remains accepted at $t = 0.001$ and $t = 0.004$. The independent BVP representation reproduces the zero-to-two transition. Figure 1 shows the shared formation bracket and the subsequent growth of the proper-area separation.

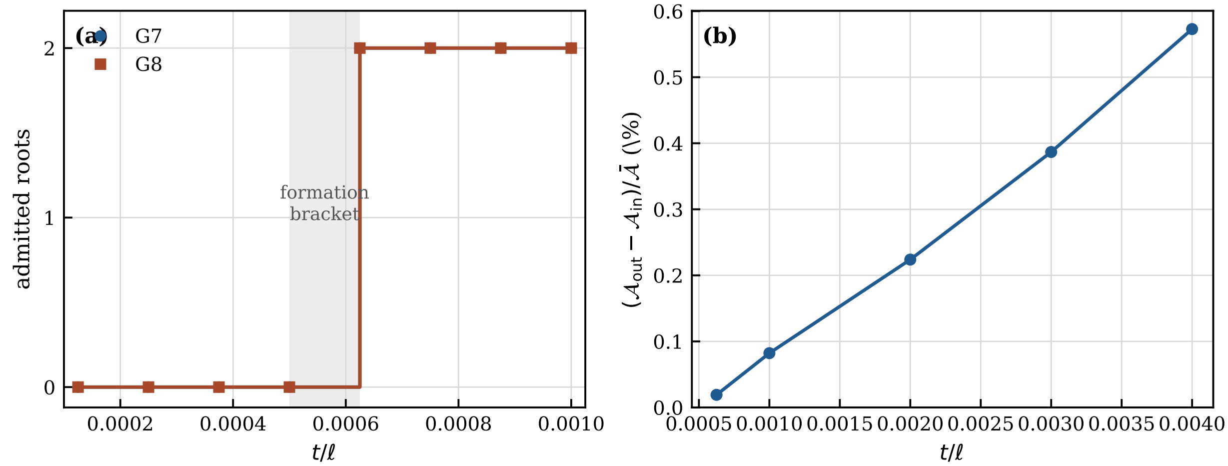

Figure 1: Historical pre-repair realization, retained for provenance and not used as independent support for the repaired result. **Left:** admitted upper-brane-capped root count on the two G7/G8 grids. Both histories change from zero to two in the common formation bracket. **Right:** one-sided proper cap-area separation on G8, shown relative to the mean area at each sampled time. The earliest positive gap is not claimed as quantitatively resolved; the growing separation on later slices is.

5.7 Historical proper-area separation

Selected and confirmation proper cap areas agree within 0.0000881%; the individual G7/G8 proper areas agree within 0.0293%. At first detection the outer-minus-inner gap is 0.02232% on G7 and 0.01908% on G8. Thus both grids reproduce the ordering, while the small gap itself differs by about 16% and is not treated as a precise near-creation observable. Thereafter the inner area decreases and the outer area increases. On G8 the gap grows to 0.5712% at $t = 0.004$, giving a progressively clearer proper-geometric separation.

5.8 Historical principal stability spectrum

Table 6 gives the principal eigenvalues evaluated from the complete evolved slice within the $SO(3)$-symmetric physical-normal perturbation sector. No nonsymmetric angular-harmonic spectrum is computed. The smallest eigenvalue-to-estimated-error margin is approximately 178.

The operator is formed at 33, 49, 65, and 81 angular nodes using centered physical-normal perturbations of relative size 10^{-5}; the 81-node matrix is repeated at 2×10^{-5}. A manufactured flat-slice control with isotropic $K^i{}_j = 0.8$ has the exact half-sphere radius $R = 1.25$ and invariant Neumann eigenvalues $\lambda_0 = -1.92$ and $\lambda_2 = 3.20$. The calculation returns -1.919593 and 3.200406, endpoint-elimination defects below 10^{-10}, and a two-step matrix difference of 1.758×10^{-6}. Every physical 49/65/81 sequence retains one sign; the largest 65/81 change is 7.54×10^{-5} and the largest step-induced eigenvalue change is 7.35×10^{-4}.

Table 6: Historical pre-repair principal MOTS-stability eigenvalues; entries are G7/G8 and are not used to close the repaired result.

t	inner branch	outer branch
0.000625	$-0.10391/-0.08791$	$+0.10879/+0.09127$
0.001	$-0.29251/-0.28800$	$+0.33742/+0.33144$
0.004	$-0.71802/-0.71689$	$+1.06253/+1.06003$

The inner branch is therefore outward-unstable and the outer branch outward-stable in the linear MOTS sense. This conclusion does not imply nonlinear basin stability.

6 Risk–criterion–outcome–consequence matrix

Table 7 collects the load-bearing validation results in the same risk–criterion–outcome–consequence order used to delimit the claim.

Table 7: Scientific validation matrix.

Risk	Criterion	Outcome	Consequence
Shared temporal-interpolation bias in the critical point	Repeat the complete G10 parent, continuation, and critical analysis at $\Delta t/2$	The $\sim 1.8 \times 10^{-7}$ brackets overlap; their midpoints differ by 9.17×10^{-10}; relative changes in area, a, and b are below 10^{-4}	Strongly constrains the prescribed temporal concern without treating the midpoint shift as uncertainty; no quarter-step extension is triggered
Systematic temporal-evolution error	Rebuild and evolve the G10 parent at $\Delta t/2$ under the same ordered physical gates	The independent half-step parent passes finite, signature, constraint, wall, axis, endpoint, and replay gates before reproducing the critical observables	Provides an indirect evolution check, not a standalone constraint-convergence study
Insufficient spacetime-field resolution	Repeated G11 trajectories identical; every successive field/tensor discrepancy decreases; tensor ratios below 0.002	The repaired-parent field/tensor comparison passes on G9/G10/G11	Addresses the field/tensor concern for the same repaired line
Unverified direct G11 MOTS observables	Exactly two branches and 1% adjacent-grid geometry transfer at a common time	At $t = 0.0015$ the repaired-parent search admits exactly two opposite-stability branches on G9/G10/G11; maximum geometry transfer 1.76×10^{-7}	Establishes direct three-grid formation geometry and stability on the repaired parent

Risk	Criterion	Outcome	Consequence
Spectral condition gate censors the fold	Keep the historical spectral detector distinct from the repaired BVP and augmented corrector	The 10^5 gate is never applied to the repaired search or continuation; acceptance instead uses BVP residual, endpoint, domain, and cross-evaluator gates	Removes the proposed censoring mechanism; no condition-number trend is claimed because none was archived
Boundary discretization changes the critical mode	Repeat angular resolution, Fréchet step, and the physical Robin reduction	The principal sign is stable over 49/65/81 nodes and step doubling; the manufactured free-boundary control passes	Supports the principal mode under the implemented seven-point boundary treatment; no alternative stencil width is claimed
Pair anatomy does not identify a fold	Continue both branches through a principal zero and isolate the near-critical mode	The continuation encloses the zero on all grids; the near-critical mode is sign-definite and the next reduced-sector eigenvalue remains near 5.566	Supports a simple critical kernel rather than only a zero-to-two history
Degenerate or tangential critical point	Nonzero, step-stable near-critical discrete-adjoint coefficients with consistent grid transfer	$a \simeq -20.765$ and $b \simeq 2.808$ retain sign under step halving and across grids	Resolves temporal transversality and quadratic nondegeneracy under the stated discrete convention
Normalization makes coefficient values ambiguous	State the discrete phase, norm, adjoint, and overlap conventions	f_R has positive-real pivot and unit ℓ_∞ norm; the Euclidean left mode is scaled afterward to $f_L^* f_R = 1$	Makes the discrete values reproducible; finite nonzero rescaling cannot alter whether a or b vanishes
Restricted computation misses a nonsymmetric principal instability	Use equivariance together with positivity and uniqueness of the elliptic principal mode	The normalized principal eigenfunction must be $SO(3)$-invariant	Captures the full principal eigenvalue and simple kernel; the nonprincipal nonsymmetric spectrum remains uncomputed
Coordinate branch separation	Invariant area difference follows a one-half power with fit sensitivity	Spatial free fits give $p \simeq 0.496$; half-step G10 gives 0.4981; leave-one-out ranges stay 0.4919–0.4996	Supplies an invariant saddle-node scaling test without treating fit digits as uncertainty
Three-grid agreement is nonmonotone	Report scatter and withhold a Richardson extrapolation	Relative spreads are 7.87×10^{-6} in t_* and 2.10×10^{-6} in area; no real three-level order is defined	Supports finite-resolution consistency, not a continuum order or theorem

7 Historical pre-repair studies

The studies in this section were performed on the superseded pre-repair G7/G8 trajectory. They are preserved as part of the research record but are not transferred as controls on the repaired G9/G10/G11 saddle-node. The direct half-timestep and three-grid controls reported above are the load-bearing tests.

Within that historical trajectory, the canonical and independent wall-owner evolution paths each pass the fixed 8/16/32-step temporal sequence. Two live-gauge variations—a slow/soft and a fast/strong variant—reproduce the formation bracket and persistent pair on both G7/G8 grids. These historical studies are not used to assert repaired-line gauge robustness or foliation invariance. The load-bearing G10 critical calculation instead passes the complete half-timestep comparison in Table 5.

Historically, pair existence persisted on R8, R10, and R12. All five R8 amplitudes in $A = 7.84$–7.92 start empty and later contain the pair; R12 anchors at $A = 7.84$, 7.90, and 7.92 preserve late pair existence. The sample is not converted into a derivative, critical amplitude, or open-basin theorem.

The historical adverse-topology program includes opposite-side caps, closed-star-shaped candidates, and interval-spanning candidates. No admitted closed-star-shaped or spanning competitor is found in 480 sealed trials. The proper statement is bounded searched-class selection, not a global topology theorem.

8 Earlier G11 field closure and pre-formation search

Before the later formation and continuation calculations, the canonical spatial test compared the common-time G10–G11 discrepancy at $t = 0.001$ with the previously sealed G9–G10 discrepancy. Two full G11 trajectories are required to agree bitwise. Every load-bearing field sequence must decrease, no auxiliary RMS plateau may remain, and the relative physical metric-increment and ADM-K differences must each be below 2×10^{-3}. This test acts on evolved spacetime fields and physical tensors. By itself it did not recompute the MOTS observables on G11; that limitation is subsequently closed at $t/\ell = 0.0015$ by the direct three-grid result above. The resulting field and tensor discrepancies are reported in Table 8.

Table 8: Same-time repaired-parent spatial discrepancies.

Quantity	G9–G10	G10–G11	change
q maximum	6.06393×10^{-7}	4.06665×10^{-7}	−32.9%
q RMS	2.17493×10^{-8}	1.17272×10^{-8}	−46.1%
v maximum	1.21292×10^{-3}	8.12627×10^{-4}	−33.0%
v RMS	4.34884×10^{-5}	2.34387×10^{-5}	−46.1%
source maximum	3.98955×10^{-11}	2.82353×10^{-11}	−29.2%
source RMS	1.67857×10^{-12}	1.18785×10^{-12}	−29.2%
metric-increment absolute	1.45619×10^{-7}	1.06995×10^{-7}	−26.5%
ADM-K absolute	6.18352×10^{-5}	4.63209×10^{-5}	−25.1%

The G10–G11 relative physical-tensor discrepancies are 2.82763×10^{-4} for the metric increment and 2.32536×10^{-4} for ADM K. Both lie well below the fixed ceiling. The two G11 trajectories and complete traces are bitwise identical; all field and tensor sequences satisfy the prescribed closure criteria with no remaining auxiliary RMS plateau.

8.1 Direct surface-observable follow-up

The fixed independent BVP detector is applied to the repaired-parent G9/G10/G11 endpoints at the same time. None of the 12 fixed seeds yields an admitted branch on any grid. Table 9 reports the best near-surface on each grid.

Table 9: Repaired-parent direct surface search at $t/\ell = 0.001$.

grid	local residual	primary residual	ρ_{axis}	ρ_{brane}
G9	0.006015	0.004050	1.20448	1.46931
G10	0.006139	0.004164	1.20455	1.47113
G11	0.006092	0.004118	1.20436	1.46632

The independently evaluated expansion is positive throughout the two-cell-interior region of each best profile. The common magnitude and shape across the three grids do not resemble a worsening-resolution instability. No comparison with the superseded G7/G8 trajectory is needed for the central inference. The later same-parent continuation establishes that these three profiles lie on the pre-formation side: the critical time is near 0.00117254, after the $t/\ell = 0.001$ search and before the accepted pair at $t/\ell = 0.0015$.

An earlier comparison is retained as a technical negative rather than silently discarded. It compared canonical acceleration against a stored zero step from an experimental second wall-owner mode, so it was not a valid canonical same-mode spatial comparison. The archived experimental state differs from the canonical state at 1,196 values, with maximum absolute difference 3.981706×10^{-4} and RMS 7.7359×10^{-7}. The corrected same-mode comparison above supplies the load-bearing spatial result.

9 Negative, failed, and incomplete studies

Scientifically material negative and incomplete studies are preserved with their consequences:

- The initial no-root result is strong numerical evidence but not a formal interval proof. Consequently, the article states only bounded search nonacceptance.
- Formation times from the historical pre-repair larger-domain parents are not transferred to the repaired line or used to infer a universal formation threshold.
- Long-time and transport-oriented diagnostics do not qualify a late-time production claim. The article ends with the short-time paired-MOTS result.
- Width and boundary variants that do not satisfy their own prerequisite or identity gates are not counted as adverse scientific outcomes.
- Arithmetic-precision and parent-qualification studies that preceded the canonical evolution establish their own bounded numerical conclusions; they do not retroactively convert this result into an event-horizon or continuum claim.

10 Reproducibility and archival mapping

The principal machine-readable aggregation is stored under the project archive identifier `Protocol 230` (archive-only saddle-node finalization, 24 August 2026) in `protocol230_result.json`. Its SHA-256 file hash is

`a54782417444dfe101b291a2b6d2594e5eb2e16f5774eab1500cf80328598156`.

It binds the immutable Protocol 229 records on G9, G10, and G11. Their abbreviated SHA-256 hashes are, respectively, `a9ef9824...28b4e`, `4b3cb413...3dab`, and `1884b005...4f40`. The preceding direct formation closure is stored as `Protocol 228`, whose result file hash is `d82000ec...3f97`.

Protocol 226 remains the earlier field/tensor closure record rather than the primary scientific result. The existing-data precision and fit audit is stored as `Protocol 231`; its result SHA-256 is

`7e2f0f30ba6e0722e3f04c637557d97e837581ea7a6a2686ef6e520c21565b18`.

The independent G10 half-timestep evolution and continuation are stored as `Protocol 232`. Its result JSON and numerical archive have SHA-256 hashes

`16eea6a208b7db8be4a3cd090778e98b18f905f00ca32cba9d534e4cd25ea522`,
`98e53d892e4732398dea40b623517b43359f4e7b7f1d735286bd388797bbd999`.

The archival package binds source manifests, immutable inputs, environment receipts, prospective gate definitions, checkpoints, repeat traces, and final artifacts. The article and this supplement report scientific outcomes; the archive provides machine-level replay evidence. The complete citable package is archived in the published Zenodo version identified by doi:10.5281/zenodo.22087164. The complete version series is identified by the concept DOI doi:10.5281/zenodo.22085054. The version-controlled Python source, frozen protocol implementations, and test suite are publicly available in the GitHub repository. The Zenodo record remains the citable archival authority for the scientific results.

11 Boundary of inference

The combined evidence does not establish:

- the absence of all marginal surfaces outside the bounded search classes;
- a measured continuum convergence order or extrapolated continuum value;
- a nonprincipal spectral-gap statement outside the $SO(3)$-symmetric perturbation sector;
- nonlinear stability or a complete basin of attraction;
- an event horizon, common horizon, throat, parent/child universe, or topology change;
- a universal formation time or critical-amplitude law; or
- late-time production evolution.

These questions require separate, prospectively defined studies. In particular, global event-horizon reconstruction would constitute a distinct paper rather than an appendix to the present quasi-local claim.

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